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Roll no: 45

Sec: C

```
import pandas as pd
import numpy as np

import os

os.getcwd()

'C:\\Users\\Kabir\\Music'

os.chdir(r"C:\\Users\\kabir\\Desktop")

data=pd.read_csv("heart.csv")

data.head()

   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak
slope  \
0   52    1   0       125   212    0        1      168      0      1.0
2
1   53    1   0       140   203    1        0      155      1      3.1
0
2   70    1   0       145   174    0        1      125      1      2.6
0
3   61    1   0       148   203    0        1      161      0      0.0
2
4   62    0   0       138   294    1        1      106      0      1.9
1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0

data.tail()

      age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang
oldpeak  \
1020   59    1   1       140   221    0        1      164      1
```

```
0.0
1021   60    1   0       125   258    0        0       141       1
2.8
1022   47    1   0       110   275    0        0       118       1
1.0
1023   50    0   0       110   254    0        0       159       0
0.0
1024   54    1   0       120   188    0        1       113       0
1.4

      slope  ca  thal  target
1020      2   0     2       1
1021      1   1     3       0
1022      1   1     2       0
1023      2   0     2       1
1024      1   1     3       0
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       1025 non-null   int64  
 1   sex       1025 non-null   int64  
 2   cp        1025 non-null   int64  
 3   trestbps  1025 non-null   int64  
 4   chol      1025 non-null   int64  
 5   fbs       1025 non-null   int64  
 6   restecg   1025 non-null   int64  
 7   thalach   1025 non-null   int64  
 8   exang     1025 non-null   int64  
 9   oldpeak   1025 non-null   float64
 10  slope     1025 non-null   int64  
 11  ca        1025 non-null   int64  
 12  thal      1025 non-null   int64  
 13  target    1025 non-null   int64  
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

```
data.shape
```

```
(1025, 14)
```

```
data.size
```

```
14350
```

```
data.ndim
```

```
2
```

```
data.isna()
```

```
        age    sex     cp  trestbps   chol    fbs  restecg  thalach
exang  \
0     False  False  False     False  False  False    False    False
False
1     False  False  False     False  False  False    False    False
False
2     False  False  False     False  False  False    False    False
False
3     False  False  False     False  False  False    False    False
False
4     False  False  False     False  False  False    False    False
False
...     ...    ...    ...       ...    ...    ...      ...      ...
...
1020  False  False  False     False  False  False    False    False
False
1021  False  False  False     False  False  False    False    False
False
1022  False  False  False     False  False  False    False    False
False
1023  False  False  False     False  False  False    False    False
False
1024  False  False  False     False  False  False    False    False
False

      oldpeak  slope     ca   thal  target
0       False  False  False  False   False
1       False  False  False  False   False
2       False  False  False  False   False
3       False  False  False  False   False
4       False  False  False  False   False
...       ...    ...    ...    ...     ...
1020    False  False  False  False   False
1021    False  False  False  False   False
1022    False  False  False  False   False
1023    False  False  False  False   False
1024    False  False  False  False   False

[1025 rows x 14 columns]
```

```
data.isna().any()
```

```
age         False
sex         False
cp          False
trestbps    False
```

```
chol         False
fbs          False
restecg      False
thalach      False
exang        False
oldpeak      False
slope        False
ca           False
thal         False
target       False
dtype: bool
```

```
data.isna().sum()
```

```
age          0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           0
thal         0
target       0
dtype: int64
```

Independent and Dependent Variables

```
x=data.drop("target", axis=1)
y=data["target"]
```

Splitting of DataSet into train and Test

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2 ,rand
om_state=42)

from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

knn=KNeighborsClassifier()

knn.fit(x_train, y_train)

KNeighborsClassifier()
```

```
y_pred2=knn.predict(x_test)

accuracy = accuracy_score(y_test, y_pred2)

accuracy
```

```
0.7317073170731707
```

Support Vector Machine

```
from sklearn import svm
svm=svm.SVC()
svm.fit(x_train, y_train)
```

```
SVC()
```

```
y_pred3=svm.predict(x_test)

accuracy_score (y_test,y_pred3)
```

```
0.6829268292682927
```